

FUNGIBLE TOKENS

Fungible tokens are a cornerstone of the value proposition of Ethereum and DeFi. Any Ethereum developer can create a token divisible to a certain decimal granularity and with units that are all identical and interchangeable. By way of example, USD is a fungible asset because one \$100 bill is equivalent to a hundred \$1 bills. As we mentioned in Chapter 3, the Ethereum blockchain token interface is ERC-20.² An interface from an application developer's perspective is the minimum required set of functionalities. When a token implements the ERC-20 interface, any application that generically handles the defined functionality can instantly and seamlessly integrate with the token. Using ERC-20 and similar interfaces, application developers can confidently support tokens that do not yet exist.

The ERC-20 interface defines the following core functionality:

- `totalSupply()` – read the token's total supply
- `balanceOf(account)` – read the balance of the token for a particular account
- `transfer(recipient address, amount)` – send “amount” tokens from the transaction sender to “recipient address”
- `transferFrom(sender address, recipient address, amount)` – send “amount” tokens from the balance of tokens held at “sender address” to “recipient address”